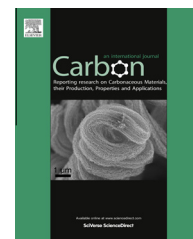


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Tuning the surface chemistry of natural graphite anode by H_3PO_4 and H_3BO_3 treatments for improving electrochemical and thermal properties

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ABSTRACT

We introduce an effective way to improve the electrochemical and thermal stabilities of natural graphite anodes of lithium ion batteries. Through a thermal decomposition of H_3PO_4 and H_3BO_3 , phosphorus and boron can be successfully incorporated into the surface of natural graphite by the formation of chemical bonds with carbon. By incorporation of these heteroatoms, the electrochemical stability of natural graphite can be significantly improved, even under severe operating conditions. From the results of various structural and electrochemical analyses, the incorporated heteroatoms are thought to be affecting the physicochemical properties of the solid electrolyte interphase (SEI) by promoting additional Li^+ consumption and thus making it more stable. In addition, we found that the heteroatoms also have potential to improve the thermal resistivity of the natural graphite anode. These findings could be helpful for developing natural graphite anodes with a good electrochemical as well as thermal properties for use in lithium ion batteries.

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1. Introduction

Increasing concerns about air pollution and depletion of fossil fuels have led to the development of electric vehicles (EVs) as a viable alternative to vehicles powered by gasoline. In comparison to other energy storage systems, lithium ion batteries are considered to be a very promising power source for hybrid electric vehicles (HEVs) and plug-in hybrid electric vehicles (PHEVs) [1,2], in which high energy and high power are needed, as well as a wider operating temperature range than in other applications [3–5]. Natural graphite (NG) has been considered to be a promising anode material for lithium ion batteries, because of its competitive cost and high specific

capacity [6–8]. Despite these features, the practical use of NG in applications of HEVs and PHEVs is still limited by its poor cyclic performance and rate capability. In addition, there is growing interest in enhancing the thermal stability of NG, because thermal runaway of lithium ion batteries is generally triggered by the anode under abnormal operating conditions, and this is directly related to the safety of lithium ion batteries. In this respect, guaranteeing the thermal stability of NG should be prioritized while maintaining good electrochemical properties.

The electrochemical performance and thermal stability of NG are highly dependent on the physicochemical properties of the solid electrolyte interphase (SEI) film, which is formed

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on the surface of NG during charge–discharge cycles. The formation of a more stable SEI film on the surface of NG, therefore, is a key challenge for ensuring the safety and improving the electrochemical performance of NG anodes for lithium ion batteries [9–12]. Until now, a number of efforts have been devoted to improve the electrochemical performance of NG, but only a few studies have been focused on improving its thermal stability. Considerable progress has been made by various surface modifications with carbons [13–16], metals [17,18], compounds [19–22], polymers [23,24], and foreign elements [25–27]. Such approaches have shown some potential for improving the cyclic performance and rate capability of NG, but further improvement is still required to meet the rigorous standards for use in EV applications. Another possible approach is to use functional additives in electrolytes, which help to form a stable SEI film on the surface of NG. However, they also have negative effects such as reducing the reversible capacity and rate capability of NG [28,29]. Finding an effective way, therefore, to improve the electrochemical and thermal stabilities of NG remains an important issue before it can be deployed in lithium ion batteries on a commercial scale.

Here, we propose an effective way to promote the formation of a more stable SEI film on the surface of NG by the incorporation of heteroatoms such as phosphorous and boron through chemical treatments with phosphoric acid (H_3PO_4) and boric acid (H_3BO_3). The incorporated heteroatoms enhance not only the thermal resistivity, but also the durability of the SEI film formed on the surface of NG, and in turn, improve the thermal stability and electrochemical performance. To verify the positive effects of this approach, we thoroughly investigated the electrochemical properties at high and low temperatures using various electrochemical and structural analyses. Our findings will provide a basis for the development of self-sustainable NG anodes for lithium ion batteries.

2. Experimental

Spherical NG particles with a size of about 7 μm (provided by Poscochemtech, Korea) were used as the raw material. P and B were separately incorporated onto the surface of NG by thermal decomposition of phosphoric acid (H_3PO_4 , Aldrich, 85% in water) and boric acid (H_3BO_3 , Aldrich, 99.9%), respectively. To introduce P onto the surface of NG, H_3PO_4 was diluted in de-ionized water and homogeneously mixed with the NG powder. The slurry was dried at 100 $^\circ\text{C}$ for 5 h in a vacuum oven to avoid the undesirable formation of a secondary phase. The collected powder was finally heated at 800 $^\circ\text{C}$ for 2 h with continuous stirring under an Ar atmosphere. For the incorporation of B, H_3BO_3 powder was dissolved in de-ionized water and thoroughly mixed with the NG powder. Then, it was dried and heated under the same conditions as for P. The amount of H_3PO_4 and H_3BO_3 was optimized to be 1 wt% of NG, as deduced in preliminary work.

The electrodes were prepared by coating Cu foil with slurries containing NG treated with H_3PO_4 or H_3BO_3 (96 wt%) as the active materials and a binder (SBR/CMC, 4 wt%) dissolved in de-ionized water. After coating, the electrode was dried at 120 $^\circ\text{C}$ for 10 h and pressed at a pressure of 200 kg/cm². For the purpose of comparison, the loading and electrode density

were fixed at 5.0 mg/cm² and 1.5 g/cc, respectively. The test cells were carefully assembled to evaluate their electrochemical performance. A porous polyethylene (PE) membrane was used as a separator and the electrolyte was 1 M LiPF_6 dissolved in a mixed solvent of ethylene carbonate (EC) and ethylmethyl carbonate (EMC) (3:7 vol, PANAX Etec Co. Ltd.). The cells were galvanostatically charged and discharged over a voltage range of 0.01–2.0 V vs. Li/Li^+ with different currents at room temperature. The electrochemical performance of the NG anode was also evaluated at a high temperature (60 $^\circ\text{C}$) and low temperature (–20 $^\circ\text{C}$) to examine its thermal stability.

The electrochemical reactions of the NG were investigated using an electrochemical voltage spectroscopy (EVS) technique to determine the Li^+ intercalation behavior under pseudo-equilibrium conditions. Three electrode cells were assembled and charged to 1.0 V vs. Li/Li^+ and aged until the voltage stabilized, as determined using electrochemical instrumentation (Wonatech, WBCS3000). After that, we recorded the cell current resulting from the application of a potential step of 10 mV until it was close to 0.02 mA and repeated this until the cell potential reached 0.01 V vs. Li/Li^+ . In addition, potentiostatic charging experiments were conducted with a BioLogic electrochemical instrument using the potentiostatic intermittent titration technique (PITT) to estimate the charge (Q) involved in electrochemical reactions. The cells were held at the open circuit voltage (OCV) until the cell potential reached equilibrium. They were then polarized to 1.0 V vs. Li/Li^+ until a low steady-state current was attained. The potential was reduced to 0.01 V vs. Li/Li^+ with a potential step of 50 mV. The corresponding anodic current was recorded over time. AC-impedance measurements were performed using a Solartron 1255 frequency analyzer combined with a Solartron 1287 ECI. AC-impedance spectra were measured by applying an AC-amplitude of 5 mV peak-to-peak over the frequency range from 10 mHz to 1 MHz.

The morphologies and microstructures of NG treated with H_3PO_4 and H_3BO_3 were examined with a field emission scanning electron microscope (FESEM, JEOL JSM-7000F) combined with energy dispersive X-ray spectroscopy (EDS). Powder X-ray diffraction (XRD) patterns of the NG were obtained using an X-ray diffractometer (EMPYREAN, PAN analytical) equipped with a 3D pixel semiconductor detector and using $\text{Cu-K}\alpha$ radiation ($\lambda = 1.54056 \text{ \AA}$). Raman spectra were collected for the NG using Raman spectroscopy (BRUKER Senterra Grating 400) with a He–Ne laser at a wavelength of 532 nm. For further characterization of the surface chemistry of NG, X-ray photoelectron spectroscopy (XPS, Thermo Scientific Sigma Probe) was employed. The thermal stability of NG was studied using differential scanning calorimetry (DSC, METTLER TOLDO STAR[®] system) after being charged to 0.01 V vs. Li/Li^+ .

3. Results and discussion

We propose an effective way for improving the electrochemical and thermal stabilities of NG by incorporation of heteroatoms into the surface of NG through the thermal decomposition of H_3PO_4 and H_3BO_3 . Fig. 1a shows XRD patterns of NG treated with H_3PO_4 (HPO-NG) and H_3BO_3

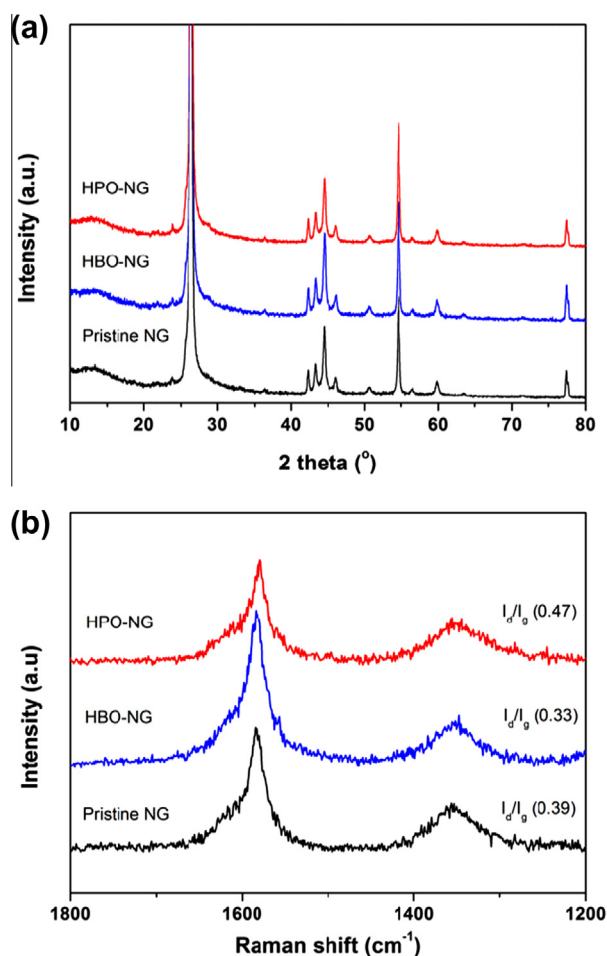


Fig. 1 – (a) Powder X-ray diffraction (XRD) patterns and (b) Raman spectra collected from pristine NG, HPO-NG, and HBO-NG.

(HBO-NG), compared with that of pristine NG. All reflections are well matched with a typical graphite structure, which is a hexagonal structure [30]. After the chemical treatments, we could not detect any evidence of the formation of impurities or a secondary phase. It is likely that H_3PO_4 and H_3BO_3 are thermally decomposed in the early stages of the heating process. P or B would be incorporated into the surface of NG in both elemental and compound forms under the given heating temperature of 800 °C.

In order to inspect structural changes in NG caused by P or B incorporation, HPO-NG and HBO-NG were characterized with Raman spectroscopy, as presented in Fig. 1b. In general, Raman signals are induced by lattice vibrations of NG, which are very sensitive to the degree of structural disorder. According to the Raman spectra, all samples have two distinctive peaks at 1360 and 1580 cm^{-1} , corresponding to the D-band and G-band of graphite structure, respectively. The G-band represents the ordered graphitic (sp^2 carbon) structure while the D-band is associated with a defective graphitic (sp^3 carbon) structure [31]. From the comparison, the intensity ratio of D-band and G-band (I_D/I_G) of pristine NG is 0.39, but this is increases slightly to 0.47 after H_3PO_4 treatment as a result of the intensity of G-band decreasing, which suggests that

partial amorphization occurred at the surface of the NG due to P incorporation [32]. In contrast, the intensity ratio decreases to 0.33 after H_3BO_3 treatment. We observed a small increase in the intensity of the G-band in HBO-NG, which is evidence for an increase in the crystallinity at the surface of NG due to B incorporation [33]. From the results, we suggest that the surface structure of NG has been effectively modified by incorporation of P and B through H_3PO_4 and H_3BO_3 treatments, respectively, and this is responsible for changing the bonding nature of C atoms at the surface of NG.

The morphologies of NG were also examined using FESEM at different magnifications, as displayed in Fig. 2. The pristine NG has a spherical shape with an averaged particle size of about 8.4 μm (Fig. 2a and b) and its surface area was measured to be 2.78 m^2/g by the Brunauer–Emmet–Teller (BET) method. Comparing this to HPO-NG (Fig. 2c and d) and HBO-NG (Fig. 2e and f), there is no significant morphological change and only small variations in surface area are observed after the H_3PO_4 (2.93 m^2/g) and H_3BO_3 (2.79 m^2/g) treatments, which might be due to structural changes induced by the incorporation of P or B at the surface of the NG.

The surface chemistry of NG was further characterized using XPS and the collected data are shown in Fig. 3. We clearly observed P and B signals on the surface of NG after the H_3PO_4 and H_3BO_3 treatments, respectively. The results indicate that P and B were successfully introduced onto the surface of NG by thermal decomposition of H_3PO_4 and H_3BO_3 at 800 °C. The deconvolution of P 2p and B 1s was carefully performed by the charging correction method based on the C 1s excitation at 284.5 eV. With regard to HPO-NG, the C 1s spectrum can be deconvoluted into five components, as shown in Fig. 3a. We fitted a small peak at 283.2 eV (peak A) induced by the formation of carbide, but it is negligible. Peak B is a dominant peak associated with graphitic carbon, which is a general feature of a well-developed graphite structure. In addition, we found a peak attribute to the C–O–P bond at 284.7 eV (peak C) [34,35], which could be have been introduced by P incorporation as a result of the H_3PO_4 treatment. Other peaks related to C atoms in carbonyl groups and carboxyl or ether groups were also found at 286.2 eV (peak D) and 288.5 eV (peak E), respectively. In addition, there is further evidence for the formation of C–O–P bonds from the P 2p excitation seen in Fig. 3b. The peak can be deconvoluted into two main components at 134.5 and 135.8 eV. It is generally accepted that peak A originates from the metaphosphate (3^+) formed in the C–O–P linkage and peak B represents P with a pentavalent (5^+) form of oxide [35]. We cannot find any evidence for elemental P around 130.0 eV in HPO-NG. On the other hand, the HBO-NG spectrum clearly showed a peak at 282.7 eV (peak A), corresponding to the chemical bond between C and B in the C 1s spectrum (Fig. 3c), which is in good agreement with the B 1s profile. A strong peak is observed at 191.8 eV, attributable to the B^{3+} resulting from the formation of the B–O bond [36]. In addition, a broad peak also appears at 186.0 eV arising from the B–C bond, as shown in Fig. 3d [36]. From the XPS results, we found that both B–O and B–C bonds can be easily formed on the surface of NG by the H_3BO_3 treatment. We note that both P and B were incorporated into the surface of NG by forming chemical bonds with

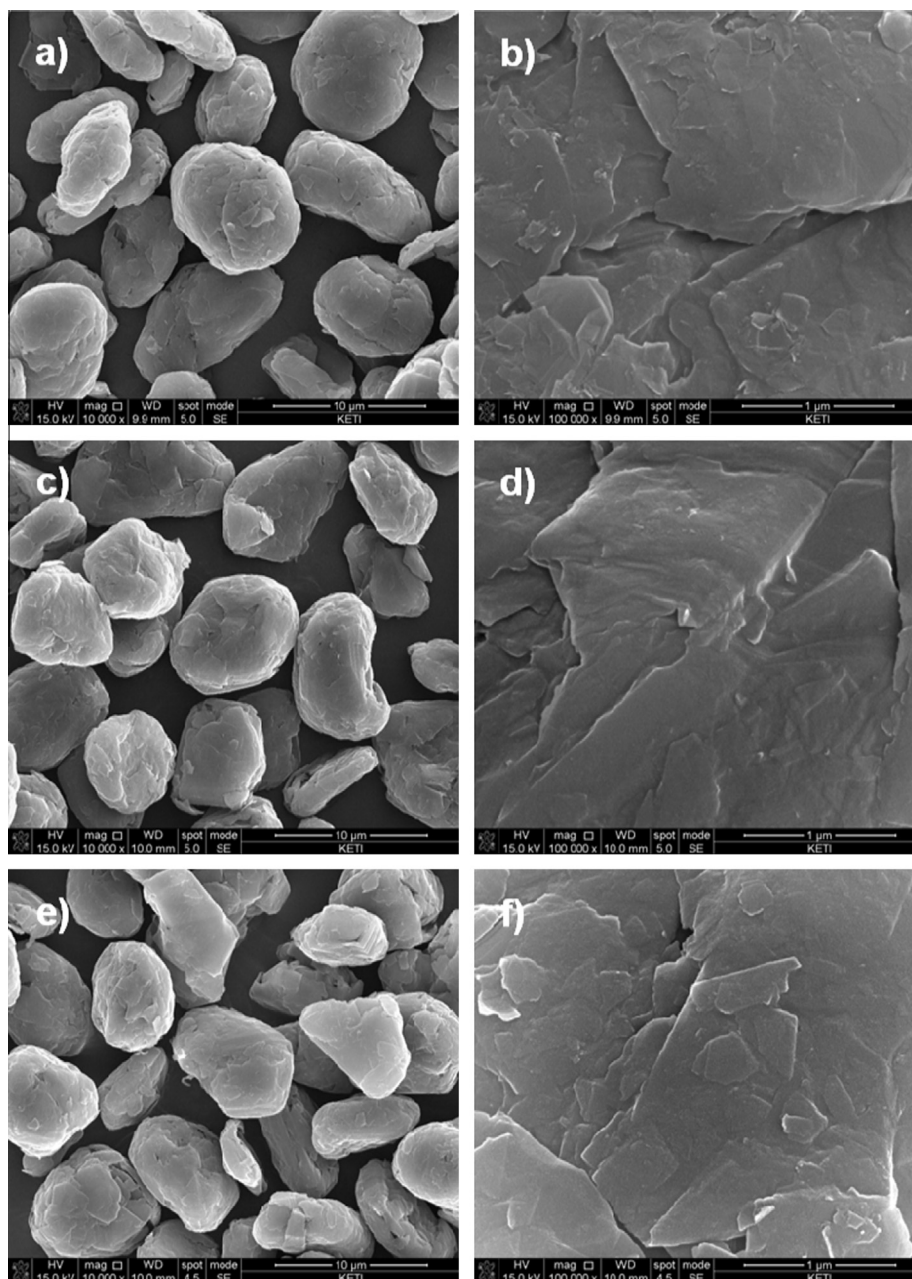


Fig. 2 – Field-emission scanning electron spectroscopy (FESEM) images for (a and b) pristine NG, (c and d) HPO-NG, and (e and f) HBO-NG at different magnifications.

C and this is responsible for changing the nature of the surface of NG.

Fig. 4a shows galvanostatic charge and discharge profiles of NG treated with H_3PO_4 and H_3BO_3 . There is no notable change in the profiles before and after the treatments. As summarized in Table 1, pristine NG exhibits a discharge capacity of 352.6 mAh/g with an initial coulombic efficiency of 94.5%. Interestingly, HPO-NG and HBO-NG also exhibit similar discharge capacities (inset), but we found a small reduction in their initial coulombic efficiencies, measured to be 92.5% and 93.2%, respectively. This originates from the increase in their charge capacities after the incorporation of P or B. This could be explained by additional Li^+ consumption

in the formation of the SEI during the first charge, because the irreversible Li^+ loss of NG mainly comes from the formation of the SEI film on its surface [37,38]. The cyclic performance and coulombic efficiency of NG, HPO-NG, and HBO-NG over 50 cycles are also compared in Fig. 4b. As can be seen, HPO-NG and HBO-NG exhibit better cyclic retention and coulombic efficiencies (inset) compared with pristine NG. HPO-NG delivers a capacity retention of 90.7% after 50 cycles, which is better than that of pristine NG (84.3%). HPO-NG also shows much higher cyclic retention of 93.6%. Furthermore, the incorporation of P or B also effectively improves the rate capability of NG. The capacity fading of NG is reduced effectively, even at a high current density of 5 C (1800 mA/g)

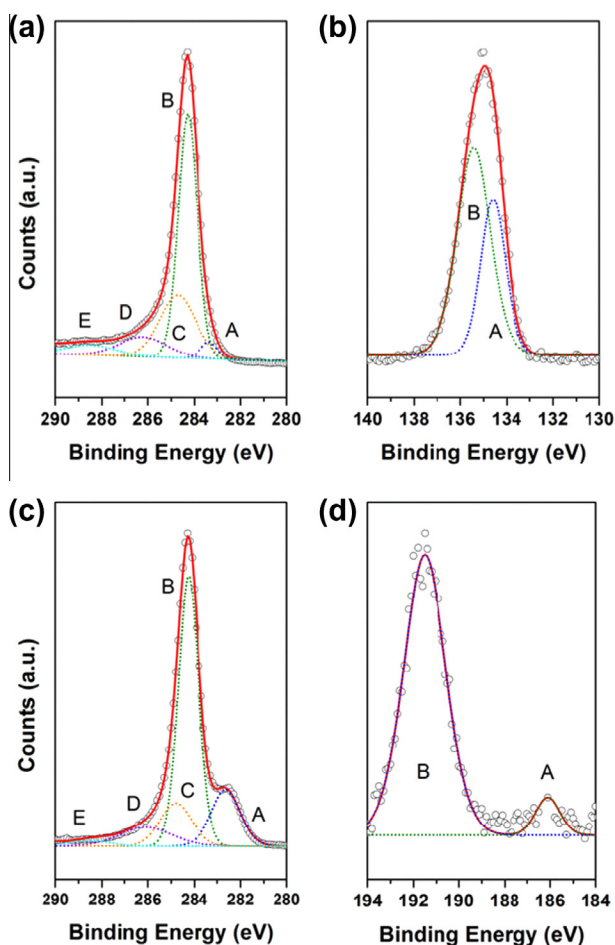


Fig. 3 – X-ray photoelectron spectroscopy (XPS) spectra of (a) C 1s and (b) P 2p signals obtained from HPO-NG; (c) C 1s and (d) B 1s signals obtained from HBO-NG.

after the H_3PO_4 and H_3BO_3 treatments, as shown in Fig. 4c. These results support the idea that the durability of the SEI film is enhanced by the incorporation of P or B, thus leading to better cyclic performance and rate capabilities. For comparative purposes, we prepared NG-800 by heating pristine NG at 800°C after mixing it with de-ionized water without any chemical source. The NG-800 shows a similar electrochemical performance to pristine NG. This is further evidence that the incorporation of P or B affects the electrochemical properties of NG.

To clarify the reason for the small decrease in the initial coulombic efficiencies of HPO-NG and HBO-NG, we inspected the electrochemical behaviors of Li^+ under pseudo-equilibrium conditions during the first charge to 0.01 V vs. Li/Li^+ using EVS technique [39], as shown in Fig. 5a. The SEI film is formed on the surface of NG in the potential range of 0.65 – 0.90 V vs. Li/Li^+ , in which the electrolyte decomposes in our system [40]. We confirmed that HPO-NG and HBO-NG exhibit distinctive potentials for the formation of SEI film. The SEI film forms at a lower potential on HBO-NG than on pristine NG, whereas for HPO-NG the difference is only slight. The EVS results indicate that the SEI formation potential could be influenced by the nature of the surface of NG. It seems that the chemical potential of NG could be altered by the

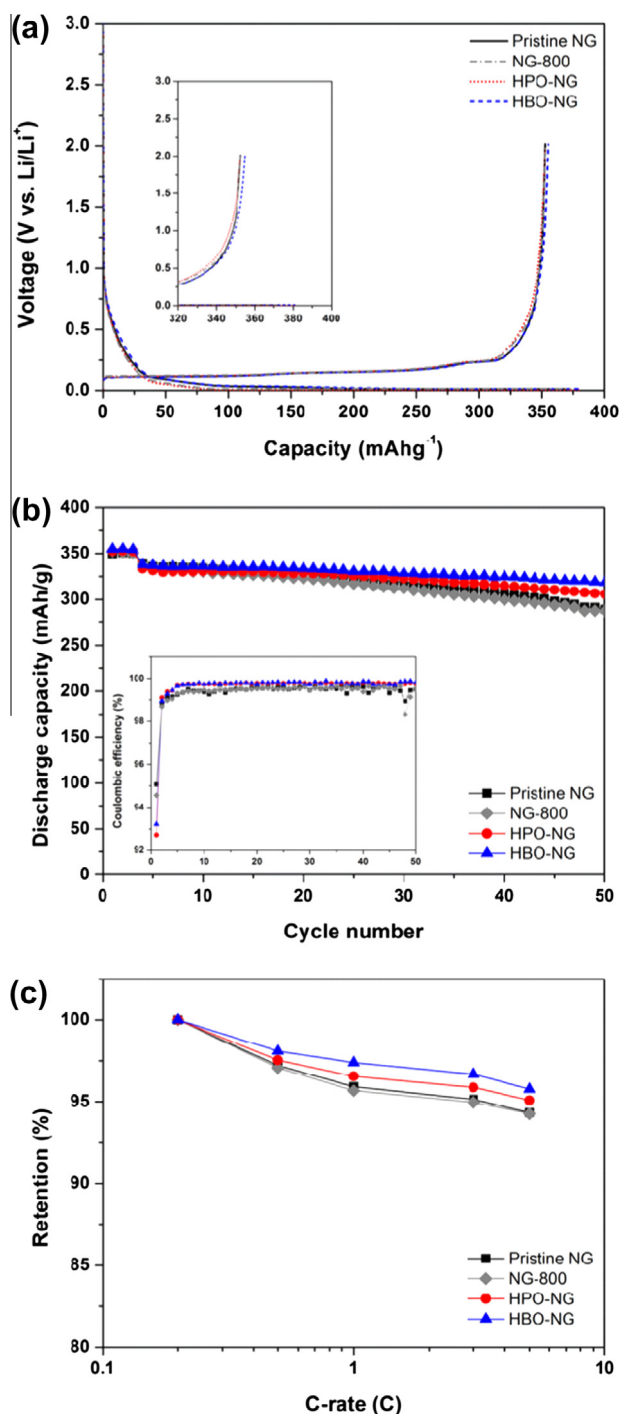


Fig. 4 – Electrochemical properties of pristine NG (■), HPO-NG (●), HBO-NG (▲), and NG-800 (◆): (a) galvanostatic charge and discharge profiles with a constant current of 0.2 C (72 mA/g) in the voltage range from 0.01 to 2.0 V vs. Li/Li^+ for the first cycle; (b) cyclic performance and coulombic efficiencies (inset) with a constant current of 0.5 C (180 mA/g) over 50 cycles; (c) rate capabilities with different currents of $0.2/0.5/1.0/3.0/5.0\text{ C}$ ($1\text{ C} = 360\text{ mA/g}$).

incorporation of heteroatoms. In addition, we conducted further electrochemical analyses on NG using PITT [41]. In the potential range of 0.65 – 0.90 V vs. Li/Li^+ , we measured the Li^+ consumption associated with the formation of the SEI film

Table 1 – Summary of electrochemical properties of pristine NG, HBO-NG, and HPO-NG obtained at the first cycles.

	Charge capacity (mAh/g)	Discharge capacity (mAh/g)	Initial coulombic efficiency (%)
Pristine NG	373.0	352.6	94.5
HPO-NG	381.1	352.5	92.5
HBO-NG	380.8	355.0	93.2
NG-800	372.9	352.4	94.5

on the surface of the NG. According to a comparison of the normalized coulomb (Q), the pristine NG exhibited 3.6% charge consumption in the given voltage range, as shown in Fig. 5b. However, HPO-NG and HBO-NG showed charge consumptions of 4.1% (Fig. 5c) and 4.4% (Fig. 5d), respectively. Such results support our hypothesis that more Li^+ were consumed in the SEI formation during the first charge after the H_3PO_4 and H_3BO_3 treatments. It is likely that the incorporated P or B compounds could promote the formation of a dense SEI film and make it more stable during cycling, leading to improvements of the electrochemical stability of NG.

Nyquist plots of the AC-impedance spectra obtained for the lithiated NG electrodes after the first charging to 0.01 V vs. Li/Li^+ are presented in Fig. 6a. Each of the impedance spectra consists of two separated arcs in the high-frequency range

and a straight line inclined at a constant angle to the real axis in the low-frequency range. The two-arc behavior of the impedance in Fig. 6a is consistent with the results for lithiated graphite electrodes reported previously elsewhere [42]. The first arc reflects Li^+ transport properties and capacitance of passive SEI film on the NG, and the second arc is ascribed to the interfacial charge transfer reaction combined with double-layer charging/discharging. The straight line at low frequencies is associated with semi-infinite diffusion of Li^+ in the NG (Warburg impedance). The measured impedance spectra were quantitatively analyzed using the equivalent circuit shown in Fig. 6b. In the equivalent circuit, R_s represents the un-compensated solution resistance; R_f and CPE_f , the resistance and the constant phase element for surface films; R_{ct} and CPE_{dl} , the charge-transfer resistance and the constant

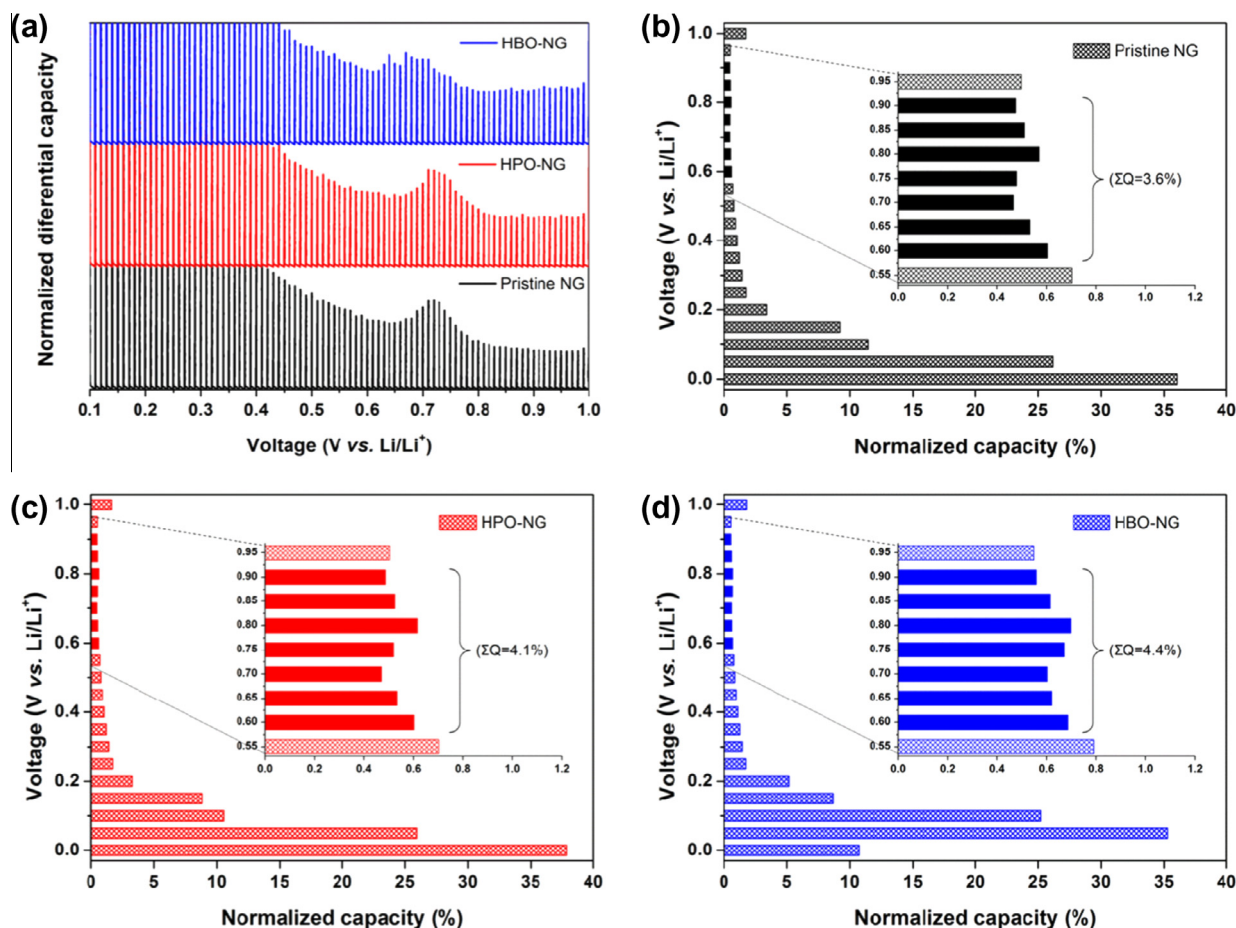


Fig. 5 – (a) Electrochemical voltage spectroscopy (EVS) profiles of pristine NG, HPO-NG, and HBO-NG and potentiostatic intermittent titration technique (PITT) results for (b) pristine NG, (c) HPO-NG, and (d) HBO-NG.

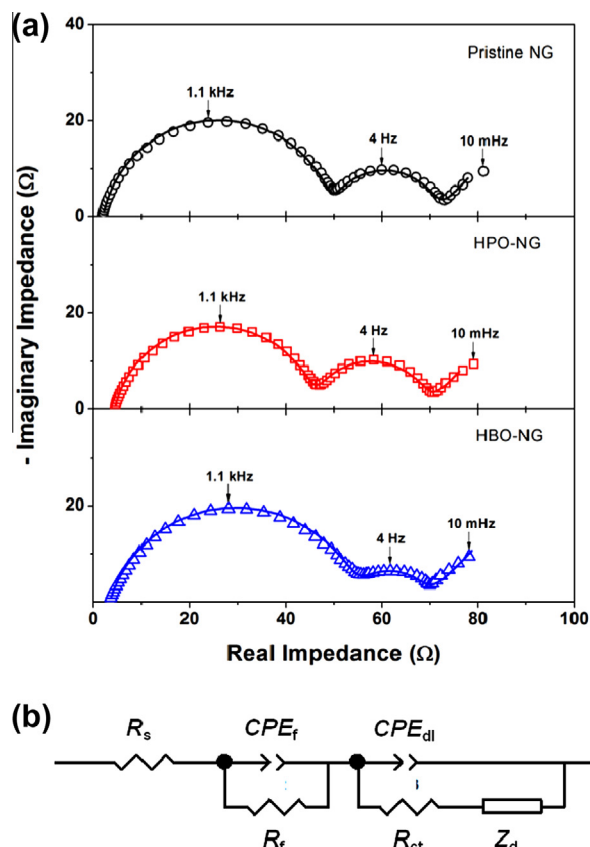


Fig. 6 – (a) AC-impedance spectra in Nyquist plots for pristine NG, HPO-NG and HBO-NG after the first lithiation (charging to 0.01 V vs. Li/Li⁺): symbols: measured, lines: fitted. **(b)** Equivalent circuit used for the CNLS fitting: R_s represents the un-compensated solution resistance; R_f and CPE_f , the resistance and the constant phase element for surface films; R_{ct} and CPE_{dl} , the charge-transfer resistance and the constant phase element for double-layer charging/discharging; and Z_d denotes the diffusion impedance.

phase element for double-layer charging/discharging; and Z_d denotes the diffusion impedance. The parameters were determined by the CNLS fitting method and listed in Table 2. It should be noted that the values of R_s and R_{ct} for the lithiated HPO-NG and HBO-NG are similar to those for the pristine NG. This confirms that the incorporation of P or B brings about no undesirable, negative effects on the Li⁺ transport kinetics across the electrode/electrolyte interface, i.e., Li⁺ migration through the SEI films and charge transfer at the interface between the SEI film and the NG.

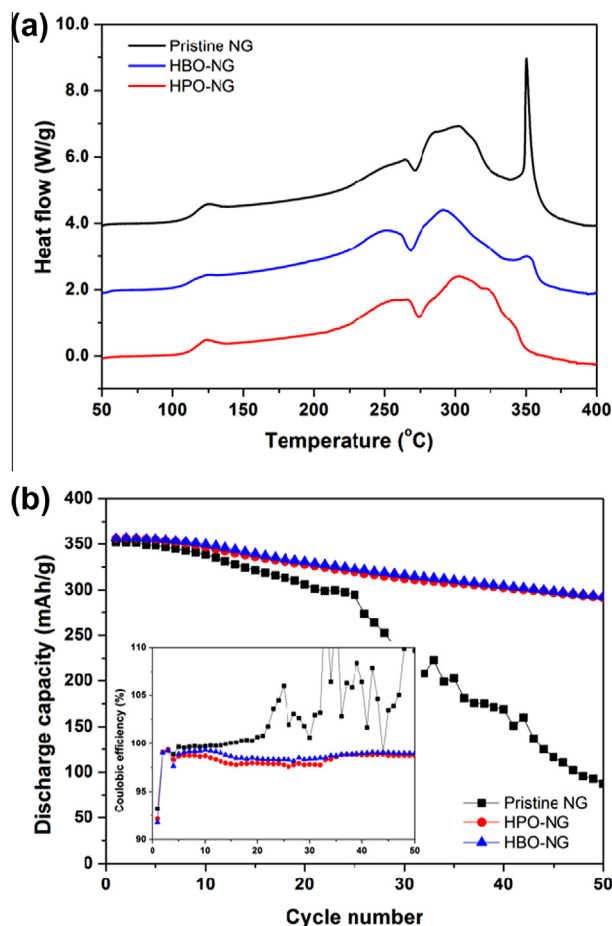


Fig. 7 – (a) Differential scanning calorimetry (DSC) profiles obtained from pristine NG, HPO-NG, and HBO-NG after charging to 0.01 V vs. Li/Li⁺; **(b)** high-temperature cyclic performance and coulombic efficiencies (inset) of pristine NG (■), HPO-NG (●), and HBO-NG (▲) with a constant current of 0.5 C (180 mA/g) at a temperature of 60 °C.

Another important issue with NG is the enhancement of its thermal stability to make it suitable for use as an anode in EV applications. We determined the thermal stability of NG before and after the H₃PO₄ and H₃BO₃ treatments. Fig. 7a shows a comparison of the DSC curves obtained from the NG after being charged to 0.01 V vs. Li/Li⁺. The measurements were conducted in the temperature range of 25–400 °C with a heating rate of 5 °C/min. It can be seen that the overall exothermic heat flow was reduced by the H₃PO₄ and H₃BO₃ treatments. In particular, an exothermic peak around 120 °C,

Table 2 – Electrical parameters of pristine NG, HPO-NG and HBO-NG determined from the CNLS fitting of the impedance spectra (Fig. 6a) to the equivalent circuit (Fig. 6b).

	R_s (Ω)	CPE_f		R_f (Ω)	CPE_{dl}		R_{ct} (Ω)	Z_d		
		C (μF s ⁿ⁻¹)	η (–)		C (mF s ⁿ⁻¹)	η (–)		R_d (Ω)	T (s)	φ (–)
Pristine NG	2.0	9.0	0.89	47.8	2.9	0.90	21.6	122.5	941.1	0.55
HPO-NG	4.4	9.6	0.88	41.7	2.4	0.87	23.3	102.2	735.8	0.54
HBO-NG	3.4	13.9	0.82	52.1	3.4	0.86	13.7	165.1	1088.0	0.52

$$CPE = C(j\omega)^n \text{ and } Z_d = R_d \coth ([jT\omega]^\phi) / ([jT\omega]^\phi).$$

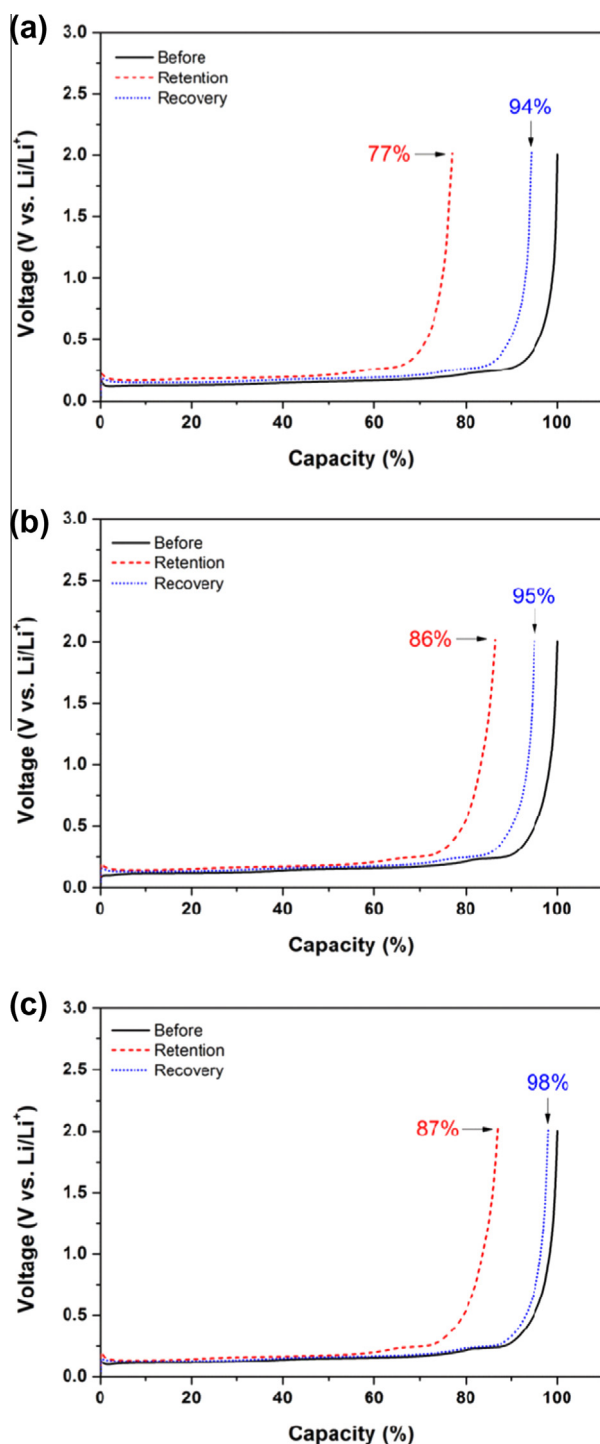


Fig. 8 – Comparisons of retention (dash-line) and recovery (dot-line) profiles of (a) pristine NG, (b) HPO-NG, and (c) HBO-NG after storage at 80 °C for 48 h.

which is mainly due to the SEI decomposition of NG, was effectively suppressed in both the HPO-NG and HBO-NG [43]. The heat flows in HPO-NG and HBO-NG were measured to be 580 and 630 W/g, respectively, which are lower than that of pristine NG with a value of 650 W/g. This reveals that the thermal resistivity of the SEI film on the surface of NG can be increased by P or B incorporation. A strong exothermic peak was observed above 350 °C in the pristine NG, and it is

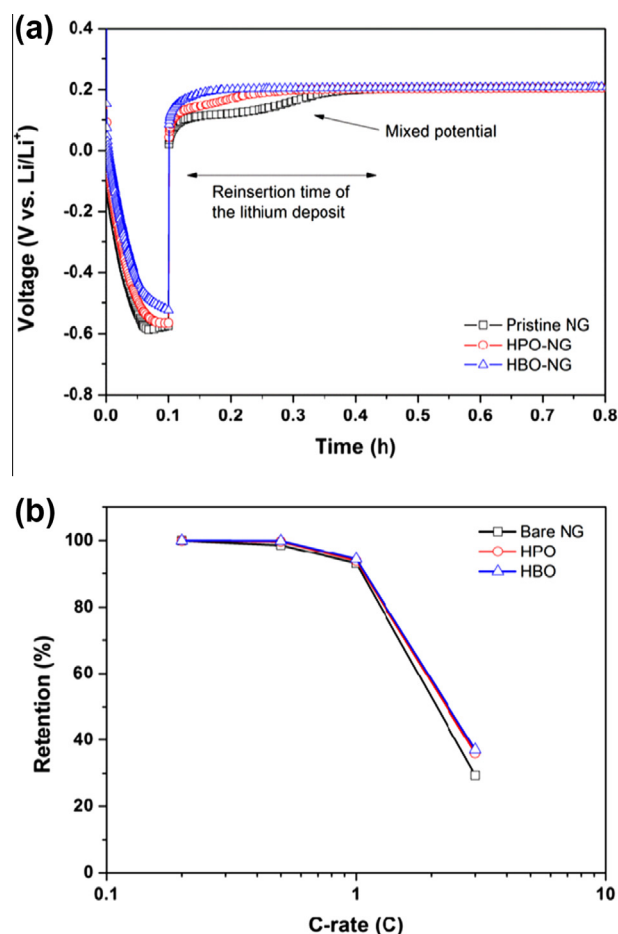


Fig. 9 – Electrochemical performance of pristine NG (□), HPO-NG (○), and HBO-NG (△) at a low temperature of −20 °C: (a) open circuit voltage (OCV) profiles during relaxation after applying a constant current of 1 C (360 mA/g) for 6 min; (b) rate capabilities with different currents of 0.2/0.5/1.0/3.0 C (1 C = 360 mA/g).

generally accepted that this arises from the structural degradation of graphite in a high-temperature environment [43]. This peak was also notably reduced in both HPO-NG and HBO-NG, which is also reliable evidence for the enhancement of the thermal stability of NG. This characteristic is directly reflected in the high-temperature cyclic performance of NG at 60 °C, as presented in Fig. 7b. After P or B incorporation, a larger discharge capacity can be retained during cycles in a high-temperature environment. Both HPO-NG and HBO-NG exhibited better capacity retention after 50 cycles with values of 84.4% and 83.2% at a constant current of 0.5 C (180 mA/g), whereas pristine NG displays a capacity retention of only 27.5% due to the significant degradation of coulombic efficiency during cycles (inset). This demonstrates that the thermal resistivity of the SEI film on the surface of the NG was effectively enhanced by the incorporation of P or B.

Additionally, we also examined the degradation of the electrochemical performance of NG after storage at high temperature. Each cell was stored in a hot box, and the temperature was maintained at 80 °C for 48 h after being charged to 10 mV. After that, the cells were discharged to 2.0 V vs.

Li/Li⁺ at room temperature to estimate the capacity retention, and subsequently, they were cycled to determine capacity recovery. As seen in Fig. 8a, the capacity loss of pristine NG after storage at 80 °C was estimated to be 23%. HPO-NG and HBO-NG retained more than 85% of their capacities, even after the storage, as shown in Fig. 8b and c. These results indicate that undesirable Li⁺ extraction from the host structure from thermal shock could be effectively mitigated by forming a more stable SEI film. Moreover, the recovered capacities of HPO-NG and HBO-NG were measured to be 98% and 95%, respectively, which are slightly higher than that of the pristine NG (94%). Thus, the structural degradation of NG due to thermal shock could be reduced by incorporation of P or B.

The Li⁺ acceptability of NG has also been investigated to determine its thermal stability at low temperatures. After storage at –20 °C for 12 h, each cell was electrochemically charged with a constant current of 1 C (360 mA/g) for 6 min and then relaxed until OCV was stabilized. In the case of the pristine NG, a notable mixed potential was observed caused by the re-intercalation of plated Li⁺ on the surface of NG as shown in Fig. 9a. For HPO-NG and HBO-NG, however, the relaxation time could be effectively shortened, which means that the interfacial resistance arising from the SEI film on the surface of the NG could be minimized by the incorporation of P or B [43]. In addition, the overpotential was also reduced when current was applied to the cells. Moreover, it should be noted that the rate capabilities of HPO-NG and HBO-NG were still comparable to that of pristine NG, as shown in Fig. 9b. Our observations indicate that the Li⁺ acceptance and rate capability of NG at low temperatures can be improved by modifying the surface of NG through the incorporation of heteroatoms by H₃PO₃ and H₃BO₃ treatments (Fig. 9).

4. Conclusion

With the intent of forming a more stable SEI film on the surface of NG, heteroatoms were incorporated into the surface by thermal decomposition of H₃PO₄ and H₃BO₃. The positive effects of P or B incorporation on the electrochemical and thermal stabilities of NG were carefully investigated. We found that the incorporated P or B compounds could effectively promote Li⁺ consumption for SEI formation on the surface of NG during cycling, leading to the formation of a more stable SEI film with better durability and thermal resistivity. As a result, the cyclic performance and rate capability of NG can be notably improved, even at high and low temperatures. Importantly, there is no severe increase in the interfacial impedance of NG due to the incorporation of P or B, demonstrated by impedance analyses. We believe that the results of our study will be helpful in the understanding of the interfacial reaction mechanism of NG and will provide further insight into the development of high-performance NG anode materials.

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REFERENCES

- [1] Armand M, Tarascon JM. Building better batteries. *Nature* 2008;451:652–7.
- [2] Wakihara M. Recent developments in lithium ion batteries. *Mater Sci Eng R* 2001;33(4):109–34.
- [3] Sato N. Thermal behavior analysis of lithium-ion batteries for electric and hybrid vehicles. *J Power Sources* 2001;99(1–2):70–7.
- [4] Kang B, Ceder G. Battery materials for ultrafast charging and discharging. *Nature* 2009;458:190–3.
- [5] Scrosati B, Hassoun J, Sun YK. Lithium-ion batteries. A look into the future. *Energy Environ Sci* 2011;4(9):3287–95.
- [6] Endo M, Kim C, Nishimura K, Fujino T, Miyashita K. Recent development of carbon materials for Li ion batteries. *Carbon* 2000;38(2):183–97.
- [7] Wu YP, Rahm E, Holze R. Carbon anode materials for lithium ion batteries. *J Power Sources* 2003;114(2):228–36.
- [8] Shim J, Striebel KA. Cycling performance of low-cost lithium ion batteries with natural graphite and LiFePO₄. *J Power Sources* 2003;119–121:955–8.
- [9] Natarajan C, Fujimoto H, Tokumitsu K, Mabuchi A, Kasuh T. Reduction of the irreversible capacity of a graphite anode by the CVD process. *Carbon* 2001;39(9):1409–13.
- [10] Yazami R. Surface chemistry and lithium storage capability of the graphite–lithium electrode. *Electrochim Acta* 1999;45(1):87–97.
- [11] Aurbach D, Markovsky B, Weissman I, Levi E, Ein-Eli Y. On the correlation between surface chemistry and performance of graphite negative electrodes for Li-ion batteries. *Electrochim Acta* 1999;45(1):67–86.
- [12] Vetter J, Novák P, Wagner MR, Veit C, Moller KC, Besenhard JO, et al. Ageing mechanisms in lithium-ion batteries. *J Power Sources* 2005;147(1–2):269–81.
- [13] Zhao H, Ren J, He X, Li J, Jiang C, Wan C. Purification and carbon-film-coating of natural graphite as anode materials for Li-ion batteries. *Electrochim Acta* 2007;52(19):6006–11.
- [14] Zhang HL, Liu SH, Li F, Bai S, Liu C, Tan J, et al. Electrochemical performance of pyrolytic carbon-coated natural graphite spheres. *Carbon* 2006;44(11):2212–8.
- [15] Sato Y, Kikuchi Y, Nakano T, Okuno G, Kobayakawa K, Kawai T, et al. Characteristics of coke carbon modified with mesophase-pitch as a negative electrode for lithium ion batteries. *J Power Sources* 1999;81–82:182–6.
- [16] Wang HY, Yoshio M, Abe T, Ogumi Z. Characterization of carbon-coated natural graphite as a lithium-ion battery anode material. *J Electrochem Soc* 2002;149(4):A499–503.
- [17] Kottogoda IRM, Kadoma Y, Ikuta H, Uchimoto Y, Wakihara M. Enhancement of rate capability in graphite anode by surface modification with zirconia. *J Electrochem Solid-State Lett* 2002;5(12):A275–8.
- [18] Kim SS, Kadoma Y, Ikuta H, Uchimoto Y, Wakihara M. Electrochemical performance of natural graphite by surface modification using aluminum. *J Electrochem Solid-State Lett* 2001;4(8):A109–12.
- [19] Huang H, Kelder EM, Schoonman J. Graphite–metal oxide composites as anode for Li-ion batteries. *J Power Sources* 2001;97–98:114–7.
- [20] Zhang SS, Xu K, Jow TR. Effect of Li₂CO₃-coating on the performance of natural graphite in Li-ion battery. *Electrochem Commun* 2003;5(12):979–82.
- [21] Wang Y, Lee JY, Chen BH. Microemulsion synthesis of tin oxide-graphite nanocomposites as negative electrode materials for lithium-ion batteries. *J Electrochem Solid-State Lett* 2003;6(1):A19–22.
- [22] Lee SE, Kim E, Cho J. Improvement of electrochemical properties of natural graphite anode materials with an ovoid

- morphology by AlPO_4 coating. *J Electrochem Solid-State Lett* 2007;10(1):A1–4.
- [23] Zhou YF, Xie S, Chen CH. Pyrolytic polyurea encapsulated natural graphite as anode material for lithium ion batteries. *Electrochim Acta* 2005;50(24):4728–35.
- [24] Tsumura T, Katanosaka A, Souma I, Ono T, Aihara Y, Kuratomi J, et al. Surface modification of natural graphite particles for lithium ion batteries. *Solid State Ionics* 2000;135(1–4):209–12.
- [25] Zhang T, Gao J, Fu LJ, Yang LC, Wu YP, Wu HQ. Natural graphite coated by Si nanoparticles as anode materials for lithium ion batteries. *J Mater Chem* 2007;17:1321–5.
- [26] Park MS, Kim JH, Jo YN, Oh SH, Kim H, Kim YJ. Incorporation of phosphorus into the surface of natural graphite anode for lithium ion batteries. *J Mater Chem* 2011;21:17960–6.
- [27] Tanaka U, Sogabe T, Sakagoshi H, Ito M, Tojo T. Anode property of boron-doped graphite materials for rechargeable lithium-ion batteries. *Carbon* 2001;39(6):931–6.
- [28] Herstedt M, Andersson AM, Rensmo H, Siegbahn H, Edström K. Characterisation of the SEI formed on natural graphite in PC-based electrolytes. *Electrochim Acta* 2004;49(27):4939–47.
- [29] Zhang SS. A review on electrolyte additives for lithium-ion batteries. *J Power Sources* 2006;162(2):1379–94.
- [30] Li ZQ, Lu CJ, Xia ZP, Zhou Y, Luo Z. X-ray diffraction patterns of graphite and turbostratic carbon. *Carbon* 2007;45(8):1686–95.
- [31] Ferrari AC. Raman spectroscopy of graphene and graphite: disorder, electron-phonon coupling, doping and nonadiabatic effects. *Solid State Commun* 2007;143(1–2):47–57.
- [32] Larrude DG, MaiadaCosta MEH, Monteiro FH, Pinto AL, Freire FL. Characterization of phosphorous-doped multiwalled carbon nanotubes. *J Appl Phys* 2012;111(6):064315–20.
- [33] Burgess JS, Acharya CK, Lizarazo J, Yancey N, Flowers B, Kwon G, et al. Boron-doped carbon powders formed at 1000 °C and one atmosphere. *Carbon* 2008;46(13):1711–7.
- [34] Lee YJ, Radovic LR. Oxidation inhibition effects of phosphorus and boron in different carbon fabrics. *Carbon* 2003;41(10):1987–97.
- [35] Puziy AM, Poddubnaya OI, Socha RP, Gurgul J, Wisniewski M. XPS and NMR studies of phosphoric acid activated carbons. *Carbon* 2008;46(15):2113–23.
- [36] Wang DW, Li F, Chen ZG, Lu GQ, Cheng HM. Synthesis and electrochemical property of boron-doped mesoporous carbon in supercapacitor. *Chem Mater* 2008;20(22):7195–200.
- [37] Wang C, Appleby AJ, Little FE. Irreversible capacities of graphite anode for lithium-ion batteries. *J Electroanal Chem* 2002;519(1–2):9–17.
- [38] Yoon S, Kim H, Oh SM. Surface modification of graphite by coke coating for reduction of initial irreversible capacity in lithium secondary batteries. *J Power Sources* 2001;94(1):68–73.
- [39] Thompson AH. Electrochemical potential spectroscopy: a new electrochemical measurement. *J Electrochem Soc* 1979;126(4):608–16.
- [40] Tang M, Miyazaki K, Abe T, Newman J. Effect of graphite orientation and lithium salt on electronic passivation of highly oriented pyrolytic graphite. *J Electrochem Soc* 2012;159(5):A634–41.
- [41] Levi MD, Wang C, Gnanaraj JS, Aurbach D. Electrochemical behavior of graphite anode at elevated temperatures in organic carbonate solutions. *J Power Sources* 2003;119–121:538–42.
- [42] Aurbach D. Review of selected electrode–solution interactions which determine the performance of Li and Li ion batteries. *J Power Sources* 2000;89:206–18.
- [43] Yamaki J, Takatsuji H, Kawamura T, Egashira M. Thermal stability of graphite anode with electrolyte in lithium-ion cells. *Solid State Ionics* 2002;148(3–4):241–5.